Thermostat State Machine — *TemperatureMachine* (Thermostat.py)

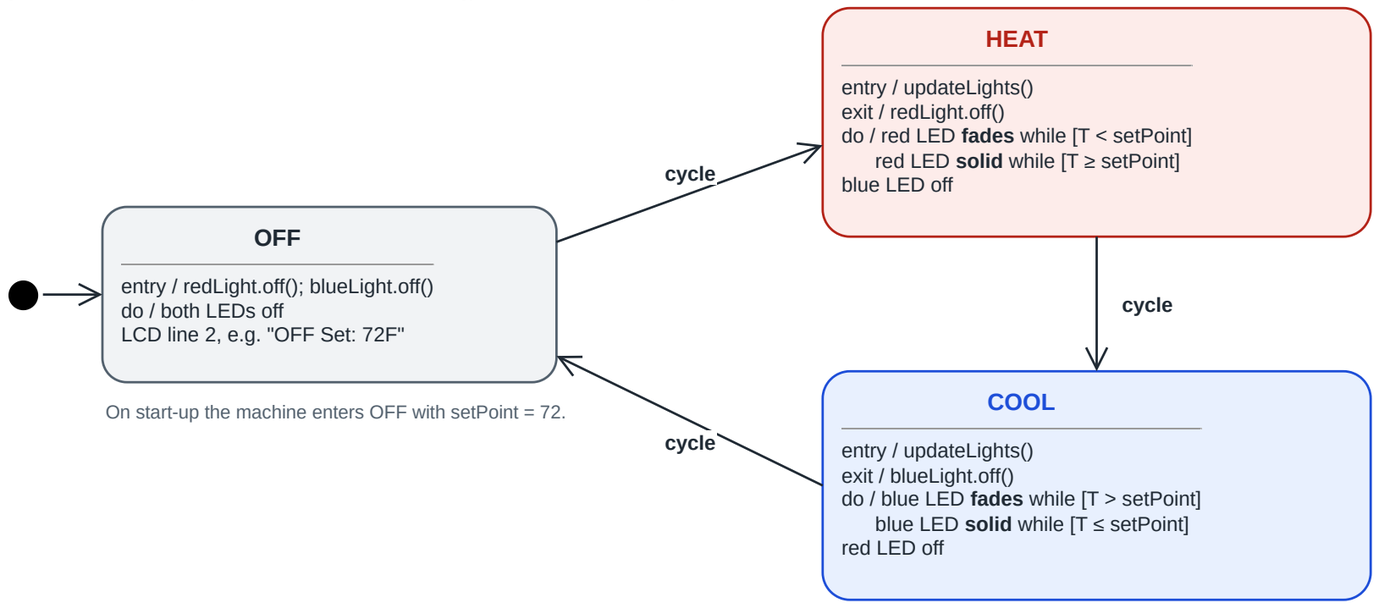
UML 2 state machine diagram · CS 350 Final Project (7-1) · Acram Kiyingi · October 2026 · python-statemachine 3.2 on Raspberry Pi 4B

TemperatureMachine

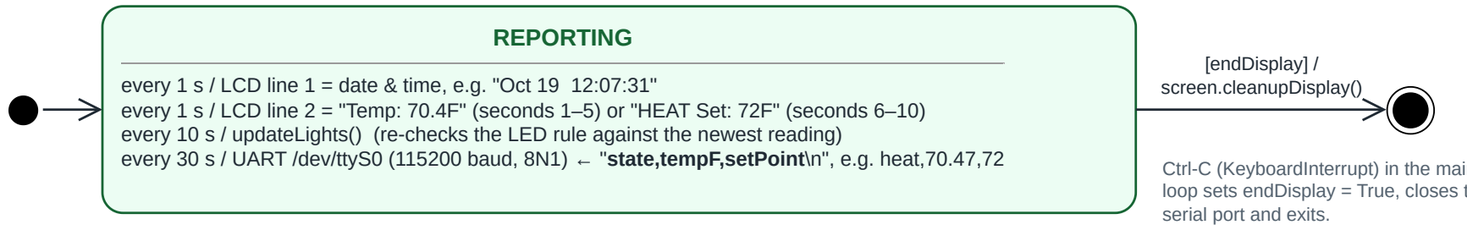
attributes: setPoint : int = 72 (°F) · ledPattern · lastFahrenheit

Internal transitions (apply in every state; the state does not change)
redButton.when_pressed (GPIO 25) / processTempIncButton(): **setPoint += 1**; updateLights()
blueButton.when_pressed (GPIO 12) / processTempDecButton(): **setPoint −= 1**; updateLights()
greenButton.when_pressed (GPIO 24) / processTempStateButton(): send("**cycle**") → drives the transitions below

Region 1 · Mode control (main thread; events arrive from gpiozero button interrupts)



Region 2 · Display & reporting (concurrent: manageMyDisplay() thread started by tsm.run(); one pass per second)



Ctrl-C (KeyboardInterrupt) in the main loop sets endDisplay = True, closes the serial port and exits.

Guard variables
T = floor(current temperature in °F). The SHT41 sensor is read over I2C (bus 1, address 0x44) under a lock; if one read fails, the last good value is used.
setPoint starts at 72 °F and moves 1 °F per button press.

LED rule — updateLights()

State	Guard	Red (GPIO 18)	Blue (GPIO 23)
OFF	—	off	off
HEAT	T < setPoint	**fade**	off
HEAT	T ≥ setPoint	**solid**	off
COOL	T > setPoint	off	**fade**
COOL	T ≤ setPoint	off	**solid**

The pattern is applied only when it changes, so a running fade is never restarted.

Hardware mapping (Raspberry Pi 4B, BCM numbering)
• Green button GPIO 24 → cycle (off → heat → cool → off)
• Red button GPIO 25 → setPoint + 1
• Blue button GPIO 12 → setPoint − 1
(gpiozero Button: internal pull-up, edge interrupt)
• LEDs: gpiozero PWMLED, red GPIO 18, blue GPIO 23
• 16×2 LCD, 4-bit: RS 17, E 27, D4–D7 = 5, 6, 13, 26
• Temperature: Adafruit SHT41 on I2C (replaces the kit AHT20)
• UART: /dev/ttyS0 (GPIO 14/15) → USB-TTL cable → /dev/ttyUSB0, read by ThermostatServer-Simulator.py

Notation (UML 2)
● initial state ◉ final state
event [guard] / action on transitions
entry / exit / do = state behaviors
dashed line = concurrent (orthogonal) regions